

CHE374 — Economic Analysis and Decision Making

Course Notes

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Fall 2026

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Week 1 - Introduction and Time Value of Money

Lecture 1: Interest Rate

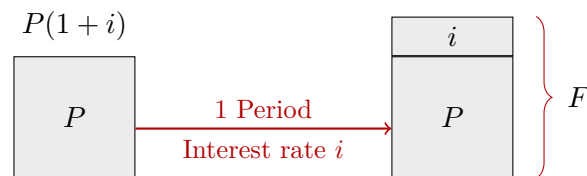
Time Value of Money

- Borrowing money has a cost for the lender, thus lending money *creates* value for the lender
- Costs and benefits occur in different time periods
- The cost/benefit today to the future is compared by **interest**

Present and Future Worth

P = present worth/**P**Principal F = future worth i = interest rate

$$F = P + Pi \quad F = P(1 + i)$$



- A higher payoff (F) is expected as i increases
- For a given F with increased i , P decreases

Interest Rate

- Expressed as %
- Vary widely over time and different types of investments.
 - A **T-Bill** (Treasury Bill, short term loan to the Bank of Canada) will earn the lowest rate of return, as it's considered "risk-free"
 - Canadian and provincial bonds pay typically higher interest rates
 - Personal/small business loans are much risky and earn significantly higher interest rates

Factors that Affect Interest Rates

- **Inflation**
- **Credit (default) Risk:** associated with the borrower being able to pay back
- **Liquidity Risk:** associated with access the invested funds during investment. Greater the risk the less liquid.
- **Maturity Risk:** associated with the time horizon. Longer the maturity, longer payoff, increases credit (and potential liquidity) risk

Example

What would the impact be expected on the interest rate associated with a large new mining project under the following scenarios, and how would the current value be impacted?

- The expected time to complete the project construction increases
Interest rate will increase as the maturity risk is directly increased, raising risks of liquidity and credit. The present value would then decrease, from both the interest rate increasing and the time horizon increasing.
- The estimate amount of ore available to be mined increases
The future cash flow would increase, thus the value of the project will increase, decreasing risk. Credit risk will drop but not others.

Summary

- **Interest** is money earned by investors for allowing others to use their money
- The **interest rate** is a rate at which interest is earned.
- The interest rate is determined by **risks** associated with the investment.

Lecture 2: Simple and Compound Interest

Present and Future Worth

P = amount of money today (principal) F_N = future amount in year N

i = interest rate I = interest amount

$$F = P + I \quad F_1 = P + Pi$$

Simple vs Compound Interest

- **Simple** applies only to the original principal (GIC)
- **Compound** interest applies to the principal AND to all interest already accrued (mortgages, credit card debt...)
- Assume compound unless stated otherwise

Simple Interest

Beginning of Period	Amount Lent	Interest Amount	Amount Owned at Period End
1	P	Pi	$P + Pi$
2	$P + Pi$	Pi	$P + 2Pi$
3	$P + 2Pi$	Pi	$P + 3Pi$
...	...	...	...
N	$P + (N - 1)Pi$	Pi	$Pi + P + NPi = P(1 + Ni)$

Compound Interest

Beginning of Period	Amount Lent	Interest Amount	Amount Owned at Period End
1	P	Pi	$P(1 + i)$
2	$P(1 + i)$	$P(1 + i)i$	$P(1 + i)^2$
3	$P(1 + i)^2$	$P(1 + i)^2i$	$P(1 + i)^3$
...	...	...	...
N	$P(1 + i)^{N-1}$	$P(1 + i)^{N-1}i$	$P(1 + i)^N$

Summary

- Simple Interest: $F_N = P + NPi$
- Compound Interest: $F_N = P(1 + i)^N$

Lecture 3: Subperiod Compounding & Effective Interest Rates

Compound Multiple Times per Year

- **r** : nominal interest rate (usually for 1 year)
- **m** : times compounded in a year, aka **subperiod**
- **i_s** : subperiod interest rate

$$i_s = \frac{r}{m}$$

Example

You lend \$100. The nominal interest rate is 6%.

What is the total amount in the account at the end of the year if the compounding period is

$$P = 100 \quad r = 6\%$$

a) 1 year:

$$\begin{aligned} F &= P(1 + r) \\ &= 100(1 + 0.06) \\ &= 106 \end{aligned}$$

b) 3 months:

$$m = 4 \quad \Rightarrow \quad i_s = \frac{r}{m} = \frac{0.06}{4} = 0.015$$

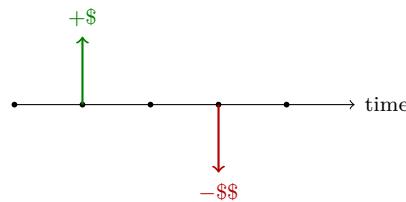
$$\begin{aligned} F &= P(1 + i_s)^m \\ &= 100(1 + 0.015)^4 \\ &= 106.14 \end{aligned}$$

Week 2 - Cash-flow Analysis

Lecture 1: Cash-flow Diagrams

Definition - Cash-flow Diagram

A simple graph summarizing the **timing** and **magnitude** of cash-flows. The x -axis is discrete time periods; the (implicit) y -axis is the size and direction of each cash-flow, drawn as a vertical arrow: an **up** arrow is a receipt (inflow, +\$), a **down** arrow is a disbursement (outflow, -\$).



Up = inflow (receipt); down = outflow (disbursement)

Definition - Cash-flow Table

A table showing the money consequences of a situation and the timing of the cash-flows: for each point in time, the activity and the resulting cash-flow. Costs/benefits occurring at the start, during, or end of a project are together called the **project life-cycle costs**.

Remark (Project Life-Cycle Cost Categories). • **Start: First (capital) cost** — expense to build/buy and install.

- **During: Revenues (sales)** — receipts from sale of products/services; **O&M costs** — expenses incurred on a regular basis (electricity, labour, repairs); **Overhaul** — a major capital expenditure partway through the asset's life.
- **End: Salvage value** — net receipt from sale/disposal at the end of the asset's *useful* life; **Scrap value** — value of the materials at the end of the asset's *physical* life; **Disposal costs** — cost to dispose of waste (removal, shipping, tipping fees, etc.).

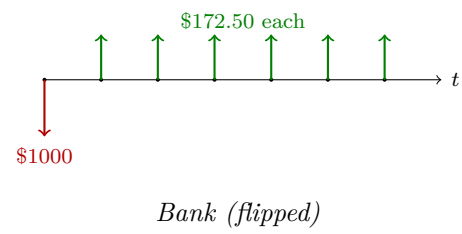
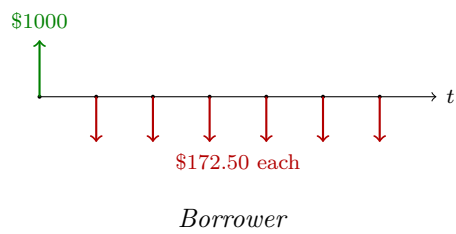
Remark (Diagram Conventions). • Time 0 is the start of period 1; the end of period N is the start of period $N + 1$.

- Unless stated otherwise: interest compounds once per period, and cash-flows occur at the **end** of the period.
- Cash-flows occurring *during* a period are summed and accounted for at the **end** of that period.

Example Loan Repayment Diagram

\$1,000 is borrowed on January 1 and repaid in 6 monthly installments of \$172.50 ($i = 1\%$ per month). From the **borrower's** perspective: an inflow of \$1,000 at $t = 0$, and outflows of \$172.50

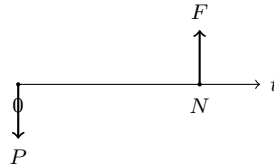
at $t = 1, \dots, 6$. From the **bank's** perspective the diagram flips: an outflow of \$1,000 at $t = 0$, and inflows of \$172.50 at $t = 1, \dots, 6$.



Lecture 2: Categories of Regular Cash-flows

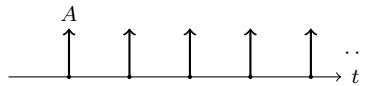
Definition - Single Payment/Receipt

A single cash-flow P (now) or F (in the future), occurring only once, N periods apart.



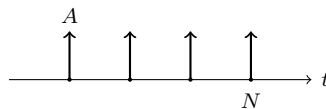
Definition - Perpetuity

A constant cash-flow A occurring every period, forever (an *infinite annuity*).



Definition - Annuity

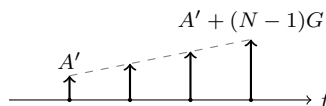
A constant cash-flow A occurring every period, for a *finite* number of periods N .



Definition - Arithmetic Gradient

A cash-flow that grows (or shrinks) by the same *amount* G each period, starting from a base A' at $t = 1$:

$$A', A' + G, A' + 2G, \dots, A' + (N - 1)G.$$

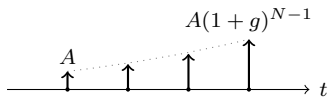


Arrow tops grow linearly (straight trend line)

Definition - Geometric Gradient

A cash-flow that grows (or shrinks) by the same *percentage* g each period, starting from A at $t = 1$:

$$A, A(1+g), A(1+g)^2, \dots, A(1+g)^{N-1}.$$



Arrow tops grow exponentially (curved trend line)

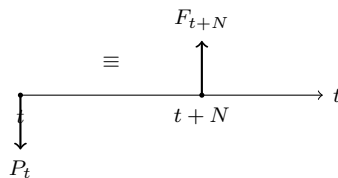
Lecture 3: Cash-flow Equivalence

Definition - Mathematical Equivalence

A consequence of the mathematical relationship between time and money. Two cash-flows P_t (at time t) and F_{t+N} (at time $t + N$) are equivalent at rate i if

$$F_{t+N} = P_t(1+i)^N \iff P_t = \frac{F_{t+N}}{(1+i)^N}.$$

If a further cash-flow F_{t+N+M} is also equivalent to P_t , then $F_{t+N+M} = F_{t+N}(1+i)^M$: equivalent amounts compound/discount consistently with each other.



Definition - Market Equivalence

A consequence of being able to exchange one cash-flow for another at *zero cost*. Exchanging a future cash-flow for a present one is **borrowing**; giving up a present cash-flow for a future one is **lending/investing**. In real markets, borrowing and lending rates usually differ.

Example T-bill Market Equivalence

A \$1,000 T-bill maturing in 6 months currently sells for \$990.10. The market has agreed this corresponds to an interest rate of 1% over 6 months.

Definition - Decisional Equivalence

Two cash-flows P_t and F_{t+N} are equivalent for a decision maker if they are *indifferent* between them. Here the interest rate is not known a priori — instead, an *implied* rate can be backed out from the decision that the two are equivalent.

Example Implied Interest Rate

A manufacturer needs 1,000 kg of aluminium on August 15 but is offered 1,100 kg on August 22 instead (same price, paid at the end of August). Accepting the delay implies an interest rate of 10% for one week.

Remark. Financial analysis uses **mathematical** and **market** equivalence to decide whether to invest.

Lecture 4: Cash-flow Factors

Remark (Standing Assumptions). Unless otherwise noted:

1. Interest compounds once per period.
2. Cash-flow occurs at the end of the period.
3. Time 0 is period 0, i.e. the start of period 1.
4. All periods are the same length.

Definition - Cash-flow Symbols

P : a single cash-flow today. F : a single cash-flow in the future. A : a finite series of equal cash-flows at regular intervals (annuity). G : a cash-flow growing by a fixed *amount* each period (arithmetic gradient) or fixed *percentage* g each period (geometric gradient, “Geom”).

Definition - Factor Notation

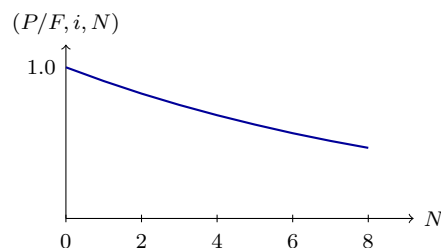
$(X/Y, i, N)$ converts a cash-flow of type Y into the equivalent amount of type X at rate i over N periods: $X = Y \cdot (X/Y, i, N)$. E.g. $F = P(F/P, i, N)$; for geometric gradients, $P = G(P/\text{Geom}, i, g, N)$.

Factor	Name
$(F/P, i, N)$	Compound amount factor
$(P/F, i, N)$	Present worth factor
$(A/F, i, N)$	Sinking fund factor
$(F/A, i, N)$	Series compound amount factor
$(A/P, i, N)$	Capital recovery factor
$(P/A, i, N)$	Series present worth factor

Remark (Relating Factors). Factors invert: since $P = A(P/A, i, N)$ and $A = P(A/P, i, N)$, we get $(A/P, i, N) = 1/(P/A, i, N)$. The same trick relates any pair of factors.

Definition - Single-Payment Factors

$$(F/P, i, N) = (1 + i)^N, \quad (P/F, i, N) = (1 + i)^{-N}.$$



$(1 + i)^{-N}$ decays toward 0 as N grows (shown for $i = 10\%$)

Example Present Value of a Perpetuity

A constant amount A paid forever has finite present value:

$$P = \frac{A}{1+i} + \frac{A}{(1+i)^2} + \cdots = \frac{A}{i}.$$

Definition - Present Worth of an Annuity

$$(P/A, i, N) = \frac{1}{i} - \frac{1}{i(1+i)^N} = \frac{(1+i)^N - 1}{i(1+i)^N}.$$

Definition - Present Worth of an Arithmetic Gradient

For a gradient contributing $G, 2G, \dots, (N-1)G$ starting at $t = 2$ (no cash-flow at $t = 0$ or $t = 1$):

$$(P/G, i, N) = \frac{1}{i^2} \left(1 - \frac{1+iN}{(1+i)^N} \right).$$

A series with initial amount A growing by G each period splits into an annuity plus a gradient:

$$P = A(P/A, i, N) + G(P/G, i, N).$$

Remark (Sign Combinations for Arithmetic Gradients). Besides $G = 0$, four cases: $A > 0, G > 0$ (positive, increasing); $A > 0, G < 0$ (positive, decreasing); $A < 0, G > 0$ (negative, becoming less so); $A < 0, G < 0$ (negative, becoming more so).

Definition - Present Worth of a Geometric Gradient

For a series $A, A(1+g), A(1+g)^2, \dots$ starting at $t = 1$:

$$(P/\text{Geom}, i, g, N) = \frac{1 - \left(\frac{1+g}{1+i}\right)^N}{i - g} = \frac{(P/A, i_o, N)}{1+g}, \quad i_o = \frac{1+i}{1+g} - 1.$$

Remark. Regardless of the pattern, present value always follows from discounting each cash-flow individually and summing:

$$P = \sum_t \frac{F_t}{(1+i)^t}.$$

Lecture 5: Worked Examples

Example Investment Decision

An investor can buy a tract of land that will be worth \$10,000 in six years. If the required rate of return is 8%, the investor should be willing to pay $P = 10,000(P/F, 8\%, 6)$ now.

Example Mortgage/Amortization

A \$300,000 mortgage at a nominal 9% (compounded semi-annually) is amortized over 15 years. The monthly payment A solves $300,000 = A(P/A, i_s, 180)$, where i_s is the effective monthly rate equivalent to the semi-annual nominal rate.

Example Sinking Fund

A roof must be replaced in 10 years for an estimated \$28,000. The equal end-of-year deposits A needed at 8% interest solve $A = 28,000(A/F, 8\%, 10)$.

Example Retirement Planning

Starting at age 20 and retiring at 65 (45 years), the equal end-of-year savings A needed to reach \$1,000,000 at an average 7% return solve $A = 1,000,000(A/F, 7\%, 45)$; a higher rate (e.g. 10%) reduces the required A .

Example Uneven Annuity Segments

To fund withdrawals of \$2,000/year for 5 years followed by \$3,000/year for the next 3 years at 8% interest, the deposit needed now is

$$P = 2,000(P/A, 8\%, 5) + 3,000(P/A, 8\%, 3)(P/F, 8\%, 5),$$

discounting the second annuity back to $t = 5$ and then to $t = 0$.

Example Present Value with an Arithmetic Gradient

A machine costs \$155 to maintain this year, with maintenance costing \$35 more each subsequent year, over an 8-year life at 6% interest. The present value of the maintenance cash-flow is

$$P = 155(P/A, 6\%, 8) + 35(P/G, 6\%, 8).$$

Adding an up-front overhaul cost simply adds that amount at $t = 0$.

Example Future Value with Periodic Dividends

\$100,000 is invested at 15% (compounded annually, not paid out), with a \$5,000 dividend received every third year for 15 years and reinvested at the same 15%. The total future value combines the compounded principal with the compounded value of each dividend.